

Q. 3	Solve Any Two of the following.		12
A)	Show that the given grammar is ambiguous grammar. $E \rightarrow E+E$ $E \rightarrow E^*E$ $E \rightarrow a$	Apply	6
B)	Explain Chomsky classification of grammar.	Remember	6
C)	Find Context Free Grammar without ϵ- productions equivalent to the following grammar . $S \rightarrow ABaC$ $A \rightarrow BC$ $B \rightarrow b \mid \epsilon$ $C \rightarrow D \mid \epsilon$ $D \rightarrow d$	Apply	6
Q.4	Solve Any Two of the following.		12
A)	Reduce the following grammar to Chomsky Normal Form(CNF). $S \rightarrow aAD$ https://www.batuonline.com $A \rightarrow aB \mid bAB$ $B \rightarrow b$ $D \rightarrow d$	Apply	6
B)	Design Push down Automata (PDA) to accept language $L=(a,b)^*$ where $n_a = n_b$.	Design	6
C)	Distinguish between Deterministic and Non Deterministic PDA.	Remember	6
Q. 5	Solve Any Two of the following.		12
A)	What are the different components of Turing machine?	Remember	6
B)	What is halt state of Turing machine? Explain Church Turing thesis.	Understand	6
C)	Design Turing machine that erases all non-blank symbols from its tape .	Design	6
	*** End ***		